

Assignment – Module 3: Prompt Engineering Fundamentals

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Section	A
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Subject	Generative AI
Submission	Handwritten copy

*Note for writing: copy the text in boxes/quotes as the “prompt”. Labels in **bold** (Instruction, Context, etc.) can be underlined in your handwritten copy.*

Q1. Improve a Bad Prompt

Bad prompt: “Explain Python.”

Improved prompt (with labels):

[Instruction] Explain what Python is and why it is popular, and show how to write a simple program in it.

[Context] I am a first-year B.Tech student with no programming experience. I am learning Python for my college course and future placements.

[Input Data] Use this example program in your explanation:
`name = input("Enter your name: ")
print("Hello", name)`

[Output Indicator] Give the answer in 5 short bullet points, followed by a line-by-line explanation of the example program. Keep it under 200 words and use simple English.

The four parts:

1. **Instruction** – tells the AI **what to do**.
 2. **Context** – tells the AI **who I am and why** I need it.
 3. **Input Data** – the **material** the AI should work on.
 4. **Output Indicator** – the **format and length** of the answer.
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Q2. Create a CRAFT Prompt

Topic: Recursion in programming

[C – Context] I am a B.Tech student preparing for my Programming semester exam next week. I understand loops and functions but find recursion confusing.

[R – Role] Act as an experienced and patient computer science teacher.

[A – Action] Explain what recursion is, how a recursive function works (base case and recursive case), and trace the factorial of 4 step by step.

[F – Format] Use: (1) a 2-line definition, (2) a real-life example, (3) a short Python code example, (4) a step-by-step trace table, (5) three common mistakes, (6) two exam-style questions.

[T – Tone] Simple, friendly and encouraging, like a teacher helping a student before an exam.

Parts identified:

- **C (Context):** student, exam next week, confused about recursion
 - **R (Role):** experienced CS teacher
 - **A (Action):** explain recursion and trace factorial(4)
 - **F (Format):** definition, example, code, trace table, mistakes, questions
 - **T (Tone):** simple, friendly, encouraging
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Q3. Demonstrate Hallucination Prevention

1. Fictional Document

Product: “SmartBottle Z1” by AquaTech (fictional)

Section 1 – Overview: SmartBottle Z1 is a smart water bottle that reminds users to drink water. It glows blue every 45 minutes.

Section 2 – Specifications: Capacity: 750 ml. Material: stainless steel. Battery: lasts 20 days on one charge. Charging: USB-C.

Section 3 – App: Works with the “AquaTrack” app on Android and iOS. The app shows daily water intake.

Section 4 – Price: ₹1,999. Available in black, white and blue.

2. Original prompt / questions

“Answer these questions about SmartBottle Z1:

Q1. Is SmartBottle Z1 dishwasher safe?

Q2. How long does the battery last?”

*(Possible problem: For Q1, the AI may guess “Yes, it is dishwasher safe”, which is a **hallucination**, because the document never says this.)*

3. Improved prompt

“Use **only** the document given below to answer the questions.

Do **not** use any outside knowledge or make guesses.

For each answer, mention the **section number** that supports it.

If the answer is not in the document, reply exactly: **"Not available in the supplied material."**

Document: [SmartBottle Z1 document]

Q1. Is SmartBottle Z1 dishwasher safe?

Q2. How long does the battery last?"

4. AI's final responses

Q1: Not available in the supplied material.

Q2: The battery lasts **20 days** on one charge. (*Supporting section: Section 2 – Specifications*)

Result: The improved prompt stopped the AI from guessing, so it gave only true, source-based answers.

Q4. Good Prompt vs. Bad Prompt

Bad prompt: "Fix my code."

Good prompt:

"I am a beginner learning Python. The function below should return the **average of a list of marks**, but it gives **ZeroDivisionError** when the list is empty.

```
def average(marks):  
    return sum(marks) / len(marks)  
print(average([]))
```

Goal: Find the cause of the error and fix it so the function returns 0 for an empty list.

Constraints: Use Python 3. Do not use any external libraries. Keep the function name the same. Make only the smallest change needed.

Output: First explain the cause in 2 lines, then give the corrected code, then one test example."

Three reasons why the new prompt is better:

1. **Gives context** – the AI knows the language, my level, the code and the **exact error**, so it does not have to guess.
 2. **Clear goal** – it says exactly what "fixed" means (return 0 for an empty list).
 3. **Has constraints and output format** – no extra libraries, a small change, and an explanation followed by code. The answer is useful and easy to understand on the first try.
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Q5. Reduce the Iteration Loop

Original prompt: “Create a portfolio website.”

CRAFT prompt:

[C – Context] I am Sagar Kumar, a first-year B.Tech (AI & ML) student at Intellipaat School of Technology, Pune. I want a personal portfolio website to share with recruiters for internships. I know basic HTML, CSS and Python.

[R – Role] Act as an expert front-end web developer and designer.

[A – Action] Create a complete, responsive one-page portfolio website using only HTML, CSS and a little JavaScript (no frameworks).

[F – Format] The website must have these sections: (1) Home with name, title and photo placeholder, (2) About Me, (3) Skills (Python, HTML, CSS, GenAI tools), (4) Projects – 3 project cards with title, description and GitHub link, (5) Education, (6) Contact form and social links. Give the code as two files: `index.html` and `style.css`, with comments. Then give 3 steps to host it free on GitHub Pages.

[T – Tone] Professional, clean and modern design, with simple, friendly text for recruiters. Colour theme: dark blue and white.

How this reduces follow-up prompts:

- **Who I am** is clear → the AI writes suitable content the first time, so I don’t have to say “make it for a student”.
- **All sections are listed** → I don’t need to ask again: “add projects” or “add contact”.
- **Technology and file format are fixed** → I don’t need to say “don’t use React” or “split into files”.
- **Tone and colours are given** → no need to ask: “make it look professional”.
- **Hosting steps are included** → no extra question afterwards.

So instead of 5–6 rounds of “change this, add that”, I get a **nearly complete website in one or two prompts**, which saves time.

— End of Module 3 Assignment —